Answer Key & Explanations

1. 2.0 Amps
    a. $R_{parallel} = (6 \times 3) / (6 + 3) = 18/9 = 2 \Omega$.
    b. $R_{total} = 4 \text{ (series)} + 2 \text{ (parallel part)} = 6 \Omega$.
    c. $I = V/R = 12V / 6 \Omega = 2A$.
2. 4.0 Volts
    a. The total current (2A) flows through the parallel bank.
    b. $V_{parallel} = I_{total} \times R_{parallel} = 2A \times 2 \Omega = 4V$. (Since B is in that parallel bank, it sees 4V).
3. 5.33 Watts
    a. $P = V^2 / R$. Voltage across C is 4V (same as B).
    b. $P = 4^2 / 3 = 16 / 3 = 5.33 W$.
4. 0.59 Ω
    a. $V_{terminal} = EMF - I(r)$.
    b. Current $I = V_{load} / R_{load} = 8.5V / 10 \Omega = 0.85A$.
    c. $8.5V = 9.0V - (0.85A \times r)$.
    d. $0.5V = 0.85r \rightarrow r \approx 0.588 \Omega$.
5. 4800 μC (or 4.8 mC)
    a. Parallel Capacitors add: $C_{total} = 100 + 300 = 400 \mu F$.
    b. $Q = C \times V = 400 \mu F \times 12V = 4800 \mu C$.
6. 12 Volts
    a. Voltage Divider Rule: $V_{R2} = V_{total} \times [R_2 / (R_1 + R_2)]$.
    b. $15V \times [4k / (1k + 4k)] = 15 \times (4/5) = 15 \times 0.8 = 12V$.
7. $0.36
    a. Energy = Power $\times$ Time = $0.1kW \times 24h = 2.4 kWh$.
    b. Cost = $2.4 kWh \times \$0.15 = \$0.36$.
8. False (Conventional flows + to -, electrons flow - to +).
9. Decreases by factor of 9 ($1/r^2$ law).
10. South Pole (Right Hand Grip Rule: Curl fingers clockwise, thumb points away from you (into the coil). The face you see is South).
11. 12 Volts
    a. Ratio $500:50$ is $10:1$ Step Down.
    b. $120V / 10 = 12V$.
12. Zero (To not impede current flow).
13. Parallel (To measure potential difference across the component).
14. Counter-Clockwise (Thumb points up, fingers curl CCW).

15. Thomas Edison
16. GFCI (Ground Fault Circuit Interrupter)
17. Higher
18. Anode (Oxidation is Loss of electrons).
19. Lose Electrons
20. South
    a. Rule: $F = qv \times B$.
    b. Velocity: East. B-Field: Down.
    c. $v \times B$ (East crossed Down) = North.
    d. However, electron is negative. Flip the direction.
    e. Force is South.
21. Variable Resistor (Potentiometer/Rheostat)
22. They all go out (Open circuit).
23. AND Gate
24. Shorted
25. SPDT (Single Pole Double Throw) - also called "Three-Way Switches" in household wiring.
26. Connected
27. Lower (The voltmeter acts as a parallel resistor, lowering equivalent resistance and thus the voltage drop at that point).
28. The longer leg is the anode (positive +), and the shorter leg is the cathode (negative -)
29. No
    a. $I = P / V = 600W / 120V = 5A$.
    b. It is exactly at the rating. Usually, fuses blow *above* the rating. It will run, but it's on the edge. (Technically "No, but it's close").
30. False (You get two smaller magnets, each with N and S).
31. C
32. B
33. D
34. B
35. D

Pembroke Hill Invitational - Repaired practice edition - Answer key

Division B | Circuit Lab | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

35. (1 points) Authored replacement for source item 35: For a simple solenoid with fixed current and fixed core geometry, does increasing the number of turns per unit length increase or decrease its magnetic field? Explain.

answer: It increases the field; in the ideal long-solenoid model B is proportional to turns per unit length times current. Holding current fixed removes the indirect effect of wire resistance. (1 points).

repair-case. (2 points) Authored written substitute for unavailable source material: Equal LED current guarantees equal brightness?

Unit: LED matching, charge-time estimates, and troubleshooting.
Worked-case data: Estimate ideal time for 1.5 Ah supplied at constant 0.30 A.

A. No

B. Always

C. Only for any red LED

D. Only at 5 V

Correct: A